

# Huawei Provider Edge Router

## (Huawei-PE-Bangalore-KA-71) - Alarm ALARM\_7501

### Resolution SOP

**Node:** Huawei-PE-Bangalore-KA-71 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM\_7501

#### Description

This Standard Operating Procedure (SOP) outlines the steps to resolve a 'ALARM\_7501' major alarm on the Huawei Provider Edge Router (Huawei-PE-Bangalore-KA-71). The alarm is triggered due to a significant issue that requires immediate attention.

#### Potential Root Causes

| Cause                                                 | Probability |
|-------------------------------------------------------|-------------|
| High CPU Utilization due to Routing Table Overflow    | Medium      |
| Misconfigured BGP Peer                                | High        |
| Hardware Fault (e.g. Fan Failure, Power Supply Issue) | Low         |

#### Diagnostic Commands

```
display cpu-usage
display routing-table
display bgp peer
display error logfile
```

#### Resolution Steps

- Step 1: {'Step': '1. Check CPU Usage and Routing Table', 'Action': "Run 'display cpu-usage' and 'display routing-table' commands to identify if the CPU is overloaded and if the routing table is overflowing."}
- Step 2: {'Step': '2. Verify BGP Peer Configuration', 'Action': "Run 'display bgp peer' command to verify the BGP peer configuration and check for any misconfigurations."}
- Step 3: {'Step': '3. Check for Hardware Faults', 'Action': "Run 'display error logfile' command to check for any hardware-related errors."}
- Step 4: {'Step': '4. Clear Routing Table and BGP Peer Sessions', 'Action': "Run 'clear routing-table' and 'clear bgp peer' commands to clear the routing table and BGP peer sessions."}
- Step 5: {'Step': '5. Verify and Configure BGP Peer', 'Action': 'Verify the BGP peer configuration and make necessary changes to resolve the issue.'}

Step 6: {'Step': '6. Restart the Router', 'Action': 'Restart the router to ensure that any configuration changes take effect.'}

## Post-Check Verification

```
display cpu-usage
display routing-table
display bgp peer
display error logfile
```